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DC Motor Intermittent Stop Diagnostic Checklist

For maintenance engineers and technicians troubleshooting PMDC and geared motors that run, stop and restart

Work through the steps in order and record every reading. Do not reset controllers or replace parts until Steps 1 and 2 are complete, because the evidence is lost once the motor restarts.

Step 1 — Record the stop pattern

- [ ] Record the running time from a cold start to the first stop, in minutes.
- [ ] Record how long the motor stays stopped and what restores running: time alone, a reset, a key or supply cycle, or nudging the shaft.
- [ ] Record what the machine was doing at the stop: ramp, lift, peak load, deceleration, reversing or end of travel.
- [ ] Note whether a welder, compressor or contactor panel switched at the same moment.
- [ ] Record ambient temperature at the motor; flag any reading above 40 °C.

Step 2 — Protection and controller checks

- [ ] Photograph the controller fault code or LED blink pattern before resetting anything.
- [ ] Record current-limit, stall-timer, under-voltage and over-voltage settings.
- [ ] Measure frame temperature immediately after a stop and compare it with the reading after 10 minutes of normal running.
- [ ] Check continuity of the thermal protector leads with the motor cold and again immediately after a stop.
- [ ] Check whether a polymer PTC resettable fuse is fitted; it resets only after power is removed.
- [ ] Measure drive heatsink temperature inside the panel and check panel ventilation for summer conditions.

Step 3 — Supply and battery checks

- [ ] Log voltage at the motor terminals and at the supply at the same time, sampling at 1 kHz or faster.
- [ ] At peak current, keep terminal voltage above the controller under-voltage threshold with margin; aim for a total drop of no more than a few per cent of nominal.
- [ ] Measure rested open-circuit voltage of each 12 V lead-acid battery: about 12.7 V when full, about 12.2 V at 50 % charge.
- [ ] Load-test each battery or compare its internal resistance with the maker's figure for a new unit.
- [ ] For lithium packs, read the BMS event log. Never lower the BMS cut-off to stop nuisance trips.
- [ ] On mains-fed drives, log DC bus voltage during deceleration and reversing to check for over-voltage trips.

Step 4 — Connections, brushes and commutator (isolate and lock out first)

- [ ] Wiggle-test each harness section, connector and gland with motor current monitored (before isolation, at light load).
- [ ] Voltage-drop test under load: no more than about 0.1 V per crimp or bolted joint, 0.2–0.3 V across a contactor.
- [ ] Inspect brush shunts for broken strands and brush springs for heat discolouration.
- [ ] Check that each brush slides freely in its holder and is longer than the manufacturer's replacement length.
- [ ] Dead-spot test: turn the shaft slowly through one revolution with a low-resistance ohmmeter across the terminals; the reading should stay steady.

- [ ] A reading that doubles or goes open at particular positions indicates an open coil, lifted riser or burnt bar.

Step 5 — Brake, interlocks and EMI

- [ ] Measure brake coil voltage at the brake terminals during release and compare it with the nameplate tolerance.
- [ ] Measure the brake air gap at three points around the circumference against the brake maker's maximum.
- [ ] With the brake released, confirm the shaft turns freely by hand.
- [ ] Trend run-enable, limit switch and E-stop inputs with time stamps; identify which signal drops first.
- [ ] Confirm suppression is fitted across relay and brake coils, and that signal cables are routed away from power cables.

Stop Pattern Quick Reference

Stop Pattern	Likely Cause	First Check
Runs 20–60 min, restarts by itself, later cycles shorter	Thermal protector cycling	Frame temperature and current at the stop
Stays stopped until supply or key is cycled	Latched controller fault or polymer PTC	Fault code; PTC in motor lead
Stops on ramps, lifts or peak loads	Supply sag, under-voltage or BMS cut-off	Terminal voltage at the stop
Stops while braking or reversing	Over-voltage from regenerated energy	DC bus voltage during braking
Stops at random when the machine moves	Loose crimp, connector or brush lead	Wiggle test; joint voltage drop
Stops at one shaft position, restarts if nudged	Commutator dead spot or stuck brush	Resistance while turning the shaft
Stops at the same point in the cycle	Load peak, brake drag or interlock	Current and PLC inputs at that point
Stops when nearby equipment switches	EMI or supply dip	Correlate stops with other equipment

Thresholds are typical field values. Always confirm limits against the motor nameplate, the controller manual and the brake or battery manufacturer's data. Never bypass a thermal protector, interlock or BMS to keep a machine running.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.